

WILMS

Women's Interest-Free Loan Management System
Implementation Guide — v1.7.3

WILMS Implementation Guide

Version: 1.7.3 Audience: Deployment teams, programme rollout leads Classification: Confidential

1. Implementation overview

This guide covers deploying WILMS for a new programme instance from infrastructure setup through go-live.
Estimated timeline: 2–4 weeks depending on data migration needs.

2. Phase 1: Infrastructure (Week 1)

2.1 Accounts

Service	Purpose
Vercel	Application hosting
Neon	PostgreSQL database
Email provider	Transactional email (optional SMS provider)

2.2 Repository setup

git clone <repository-url> cd WILMS npm ci

2.3 Environment variables

Configure in Vercel dashboard:

Variable	Value
DATABASE_URL	Neon connection string
WILMS_SESSION_SECRET	32+ character random string
NEXT_PUBLIC_API_BASE_URL	/api/wilms
NEXT_PUBLIC_USE MOCK	false

Optional: REDIS_URL for rate limiting.

2.4 Database migration

npm run db:migrate -w @wilms/domain npm run verify:migrations -w @wilms/api

2.5 Deploy

Push to production branch. Vercel auto-deploys.

Verify: GET /api/wilms/health

3. Phase 2: Organisation setup (Week 1–2)

3.1 Super Admin account

Create initial admin via seed script or first-user bootstrap. Change default password immediately.

3.2 Organisation settings

Configure: organisation name, notification preferences, quiet hours, holiday calendar.

3.3 Loan pools

Create capital pools with initial replenishment matching programme funding.

3.4 Location hierarchy

Seed Ghana locations: npm run seed:ghana-locations -w @wilms/api

4. Phase 3: User provisioning (Week 2)

4.1 Role planning

| Role | Typical count | Notes |

|-----|-----|-----|

| Super Admin | 1–2 | Programme administrators |

| Registration Officer | 2–5 | HQ registration staff |

| Collector | 10–50 | Field agents |

| Approver | 2–3 | Loan decision authority |

| Auditor | 1–2 | Read-only oversight |

4.2 Invite users

Settings !' Users !' Invite. Assign appropriate roles. Users complete first-login

4.3 Permission overrides

Grant only when role defaults insufficient. Document all overrides.

5. Phase 4: Group and collector setup (Week 2)

1. Create borrower groups aligned with community structure 2. Assign collectors to groups 3. Verify collector field shell access on mobile devices 4. Test offline payment queue

6. Phase 5: Pilot (Week 3)

6.1 Pilot scope

- 1–2 groups
- 10–20 borrowers
- 1 collector
- Full lifecycle: register ! approve ! disburse ! collect ! reconcile

6.2 Validation checklist

- ☐ Borrower registration with documents and GPS
- ☐ Loan approval workflow
- ☐ Admin fee collection
- ☐ Disbursement with pool hard-stop test
- ☐ Weekly collection (online and offline)
- ☐ Daily reconciliation
- ☐ Expense submit and review
- ☐ Report generation and export
- ☐ Notification delivery
- ☐ Audit log completeness

7. Phase 6: Go-live (Week 4)

7.1 Data migration (if applicable)

Import existing borrower/loan data via admin tools or custom migration script. Validate against source records.

7.2 Training

Distribute role manuals from documentation/user-guides/. Conduct hands-on sessions per role.

7.3 Go-live checklist

- ☐ All users invited and active
- ☐ Pools funded with correct capital
- ☐ Groups and collectors assigned
- ☐ Pilot validation passed
- ☐ Smoke tests pass: npm run smoke:production
- ☐ Ops incident process communicated
- ☐ Backup/recovery verified

8. Post go-live

Task	Frequency
Monitor health endpoint	Daily
Review reconciliation aging	Daily
Run smoke tests post-deploy	Per deploy
Export compliance pack	Monthly
Review audit log	Weekly
Update documentation	Per release

9. Rollback plan

1. Revert Vercel deployment to previous build 2. Database is forward-only — plan migrations carefully 3. Document incident in ops module 4. Communicate to users via maintenance window banner

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